

An Analysis of Purchase Order Creation Delays and Their Causes in Apparel Manufacturing Operations.

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A report submitted in partial fulfillment of the requirement for the Internship Training of the
Faculty of Science, University of Colombo

February & 2026

Acknowledgements

I wish to express my sincere gratitude to Hela Clothing Private Limited for affording me the opportunity to undertake my internship training within their operational division.

I am especially thankful to the merchandising team and my Placement Mentor, **Mr. Ranga Panditharathne** at Hela Clothing, for their guidance, support, and valuable insights into the field, which greatly assisted me in completing this study. I was better able to comprehend the practical difficulties of Purchase Order (P.O.) management because of their insights into procurement processes.

I also want to thank my Academic Mentor, **Prof. Mahendra Fernando**, for providing me with insightful academic advice on how to carry out this analytical investigation.

Lastly, I wish to express my gratitude to my colleagues and fellow interns who assisted me during my internship.

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Executive Summary

For the apparel manufacturing industry, efficient procurement activities are significant to ensure smooth operations and timely order fulfilment. One of the important operational activities in the field of supply chain management is the creation of Purchase Orders (P.O.s). Delays in creating purchase orders could have a significant impact on the business.

The objective of the study is to analyze the delay in the creation of purchase orders for Hela Apparel Holdings PLC's operations. The study has been conducted using operational data from the company's SAP system, based on the difference between the 'Purchase Order Created Date' and the 'Requirement Date.'

Statistical analysis has been performed using SPSS, and operational pattern visualization has been performed using Power BI. Business comments have been categorised based on the major reasons for delay.

Results showed that although the overall P.O creation rate appears mostly on time, further analysis revealed delays when data was segmented by material type, plant location, and customer category. This may indicate operational inefficiencies, but not across the entire system.

The findings of this study can be used to identify the major factors responsible for the delays occurring in the purchase order creation process.

1. Introduction

1.1 Background of the Project

Apparel manufacturing is a competitive global business, and speed, efficiency, and coordination are essential. The necessary materials and production resources must be available at the right time to comply with tight delivery requirements set by global buyers.

One of the important processes in production management is procurement management, specifically the generation of purchase orders for raw materials, trims, and accessories. A purchase order, abbreviated as P.O., is an official document issued by a company to its suppliers that specifies the materials needed, quantities, and delivery dates.

Delays in creating purchase orders can create several operational challenges, such as:

- Material shortages
- Production interruptions
- Increased operational costs
- Delays in final product delivery

Therefore, monitoring and improving the timeliness of purchase order creation is important for maintaining efficient manufacturing operations.

This research focuses on analysing P.O. creation delays within the operations department of Hela Clothing Private Limited, one of the leading apparel manufacturing companies in Sri Lanka.

1.2 Importance of the Study to Industry

In the apparel industry, procurement activity and production time are tightly related. Therefore, a delay in establishing the purchase order could affect several additional processes, including:

- Source of the material
- Planning for production
- Inventory control
- Scheduling shipments

Operational managers could enhance the procurement process by using the results of the study of the trends and reasons behind delays in the creation of purchase orders,

- Finding the procurement process's difficulties
- Increasing departmental coordination

- Improving correspondence with vendors
- Increasing production planning accuracy

Therefore, the company's operational performance may benefit from streamlining the purchase order creation procedure.

1.3 Objectives of the Study

1.3.1 Main Objective

To analyse purchase order creation delays and identify the major factors contributing to these delays within the operations of Hela Clothing.

1.3.2 Sub Objectives

- Measure overall P.O. delay distribution.
- Compare the delay differences among Suppliers.
- Analyse Customer level delay variations.
- To categorise business comments related to delays in major cause groups.
- To analyse whether delays vary based on material type, plant location, and customer
- Provide data-driven recommendations.

2. Methodology

2.1 Research Approach

This study applied a quantitative analytical approach to analyse the patterns and reasons behind purchase order delays in Hela Apparel's procurement operations. Quantitative approaches were chosen because the study is based on operational data generated by the enterprise and seeks to establish relationships between operational variables and purchase order creation delays.

The study is based on analysing the operational data retrieved from the company's ERP system, namely SAP. This operational data contains detailed information about procurement operations, including purchase order creation date, requirement date, plant details, customer information, and operational comments entered by the user during operations.

Statistical analysis and data visualisation techniques were applied in the study to provide both analytical and practical interpretations of the operations.

2.2 Data Source

The main data for this research were obtained from the SAP system that Hela Apparel uses for procurement and operational management. SAP contains transactional data on various procurement-related activities, including purchase order generation.

Data was obtained from various reports, including:

- Purchase Order Status Reports
- Pending Purchase Order Reports
- Order Book Data

Business comments on P.O. delays are collected via email by sending pending P.O. lists to the respective business teams.

All these reports contained historical data on various procurement activities that occurred during my internship. These reports, including the order book, were obtained over about six months, whereas the purchase orders included in these reports spanned up to 10 months.

2.3 Data Variables

Throughout the analysis, a few important variables were taken from the SAP reports. These variables have been derived based on their relevance to the research and are related to different components of the procurement process.

The key variables used in the analysis include:

Sales Order Number

- Purchase Order Created Date
- Plant Location
- Customer Name
- Material Type
- Business Comments related to delays

These factors have enabled the analysis to examine the procurement process from several angles.

2.4 Measurement of Purchase Order Delay

The primary variable used in this study is Purchase Order Creation Delay. This variable represents the difference between the date a purchase order was created and the date it was required.

The delay value was calculated using the following formula:

$$\text{P.O. delay} = \text{P.O. Created Date} - \text{Requirement Date}$$

The interpretation of this value is as follows:

- **Negative value** – Purchase order created earlier than required date
- **Zero value** – Purchase order created exactly on the requirement date
- **Positive value** – Purchase order created after the requirement date (delay)

To make analysis and visualisation easier, purchase orders were further divided into delay categories based on these delayed values.

2.5 Data Cleaning and Preparation

To ensure its accuracy and integrity, the gathered data underwent several preparation procedures before analysis.

Removing duplicate records that might have occurred from repeatedly downloading order book reports was the first step. To avoid double-counting, duplicate entries were identified by purchase order number and removed.

Missing values were then looked at. Specifically, multiple records in the dataset that lacked business comments explain the delay. Only records with available comments were included in the study on delay reasons, as these were required for the analysis of delay causes.

To ensure accurate computation of delay values, date fields were also standardised. When feasible, incomplete or inaccurate date entries were either removed from the dataset or revised.

Finally, the cleaned dataset was structured for analysis in IBM SPSS and Power BI.

2.6 Categorisation of P.O. Delays

The calculated delay days were divided into four groups based on severity to simplify the examination of Purchase Order (P.O.) creation delays. The delay was computed as the number of days between the expected P.O. creation date and the actual P.O. creation date.

The company's internal operational guidelines state that a P.O. is considered pending only if the delay exceeds 10 days. For the sake of this investigation, delays of 0 to 10 days were considered for on-time for P.O. creation.

Based on this guideline, the delay days were classified into the following categories:

- **On-Time** – Delays less than **10 days**
- **Minor Delay** – Delays between **11 and 40 days**
- **Moderate Delay** – Delays between **41 and 80 days**
- **Critical Delay** – Delays of **more than 80 days**

This categorisation enabled a structured analysis of the severity of P.O. creation delays and helped identify patterns across different customers, plants, and business comment categories. It also supported further statistical analysis and visualisation in the Power BI dashboard developed for this study.

2.7 Categorisation of Business Comments

Some business comments are recorded in the procurement system and include a brief description of the reason for the delay in the purchase order. However, these comments are mostly free-form and differ from one another.

To analyse the comments, they were manually reviewed and categorised into general operational categories. This enabled grouping comments by the reasons for the delay.

The delay categories are,

- Not Required / Small Qty
- Stock / Allocation Issues
- P.O. Already Raised Update Pending
- Delivery / Approval / Pending Issues
- System / Verification Errors

It is important to note that a large proportion of purchase orders lacked comments explaining the delay. Therefore, the analysis of delay causes was conducted only on the subset of purchase orders that included comments.

2.8 Statistical Analysis

Statistical analysis was conducted using **SPSS software** to evaluate relationships between operational variables and purchase order delays.

Two main statistical techniques were used:

2.8.1 Analysis of Variance (ANOVA)

ANOVA was used to examine whether the **average purchase order delay differs significantly across different groups**, such as plants and customers.

Two separate ANOVA tests were conducted:

- Plant-wise delay analysis
- Customer-wise delay analysis

This method allows comparison of mean delay values across multiple groups simultaneously.

2.8.2 Chi-Square Test of Independence

A Chi-Square test was conducted to examine the **association between delay categories and business comment categories**.

This test determines whether the distribution of delay reasons is independent of delay categories or whether certain operational issues are associated with specific types of delays.

2.9 Data Visualisation

In addition to statistical analysis, a Power BI dashboard has been created to display patterns in procurement performance.

The dashboard includes several visual components, such as:

- Average delay by customer
- Average delay by plant
- Average delay by purchase order creation date
- Total purchase orders by delay category
- Total purchase orders by comment category

These visualisations provide a clear view of procurement performance and help identify operational trends that may not be immediately visible through statistical tables alone.

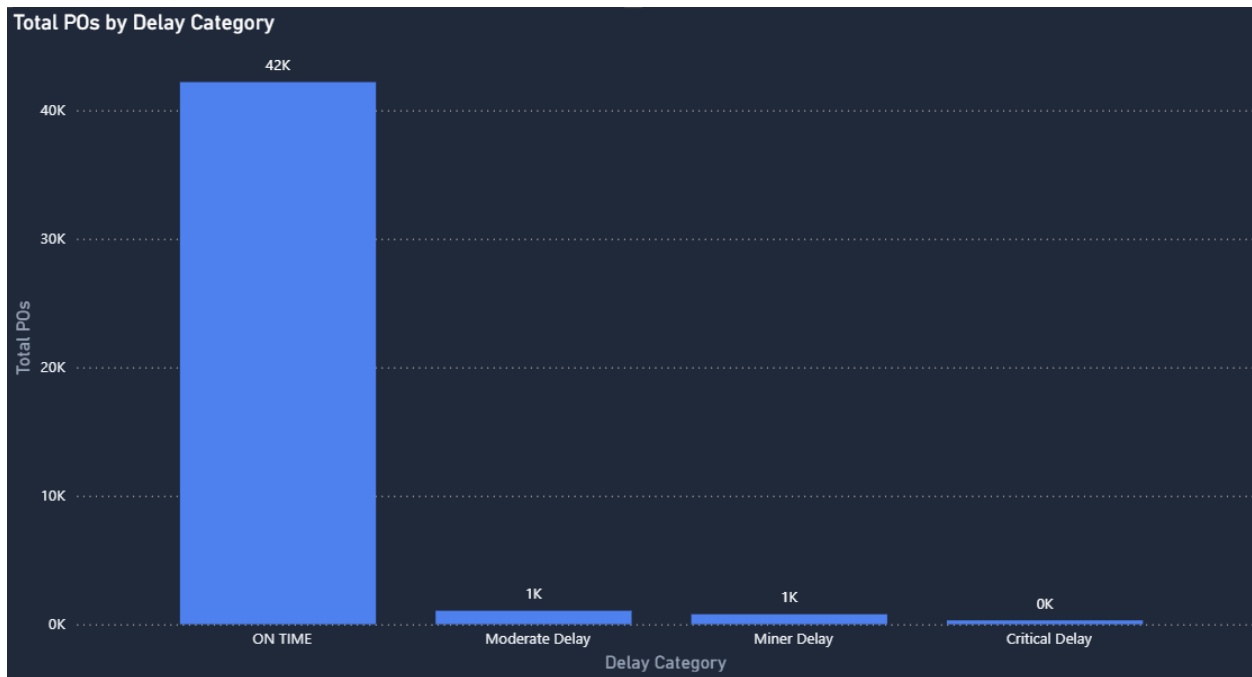
3. Results

3.1 Overview of Dataset

Purchase order records retrieved from the SAP system during the internship period comprised the dataset examined in this study. Procurement transactions involving several factories, clients, and material types are included in the data.

Understanding the overall distribution of purchase order delays was the main goal of the first investigation.

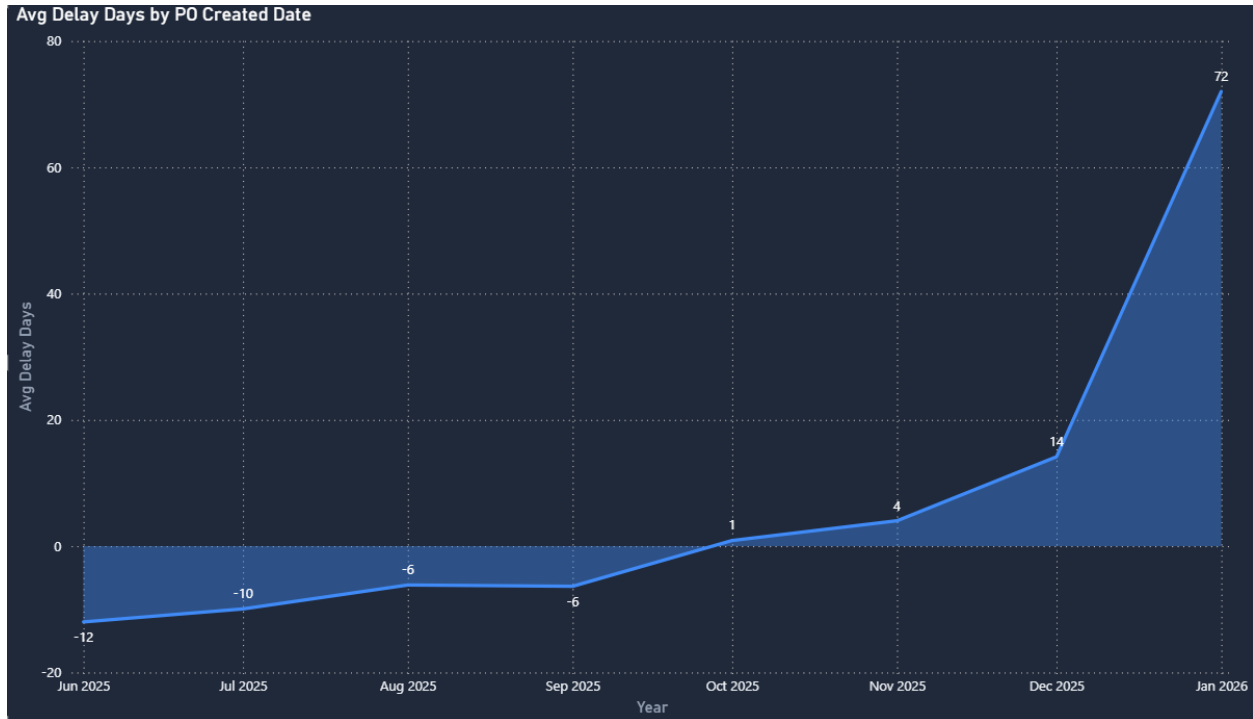
To summarise the distribution of delays, Power BI was used to produce a visualisation that displayed Total Purchase Orders by Delay Category.



The findings show that most purchase orders were made either on time or very close to the deadline. However, a significant percentage of purchase orders were delayed, suggesting inefficiencies in the procurement process.

3.2 Delay Trend Over Time

The visualisation **Average Delay by P.O. Created Date** was used to examine how procurement delays changed over time. The following figure shows the trend in Fabric Materials' delays.



From these results, we can see fluctuations in the level of delay across different time periods. The 2025 Dec – 2026 Jan period shows higher levels of delay than others.

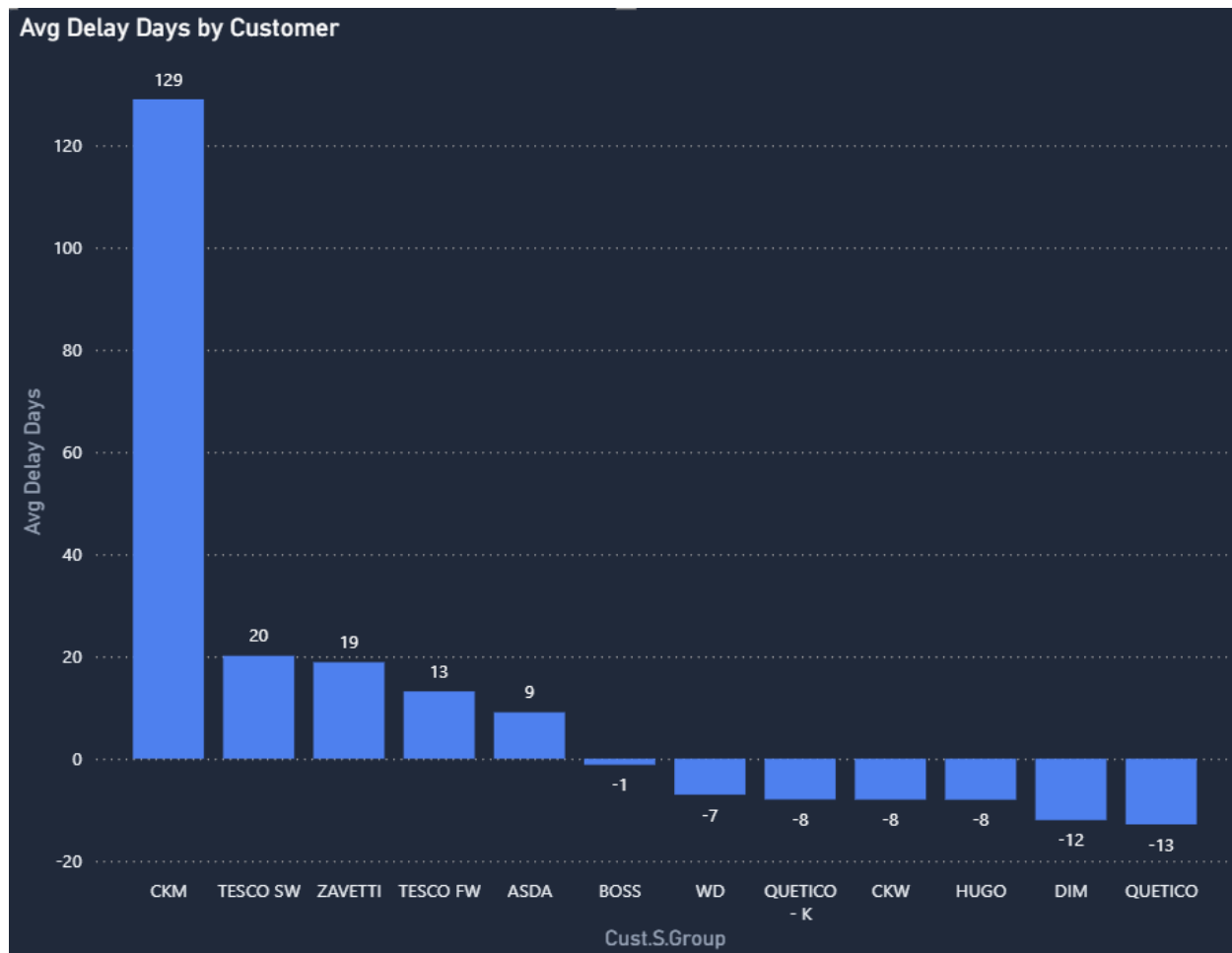
Such changes may include operational conditions like:

- Increased production requirements
- Increased workload for procurement
- Alteration of planning schedules

Such observations can help identify periods that may need improvement in procurement.

3.3 Customer-wise Delay Analysis

Customer-specific analysis was conducted to determine whether purchase order delays vary by customer.



The visualisation indicates that average delay levels of **Fabric P.O.s** differ among customers. **BOSS, WD, CKW, and HUGO** show relatively low average delays, while **CKM** shows higher delays.

These differences may be influenced by several factors, including:

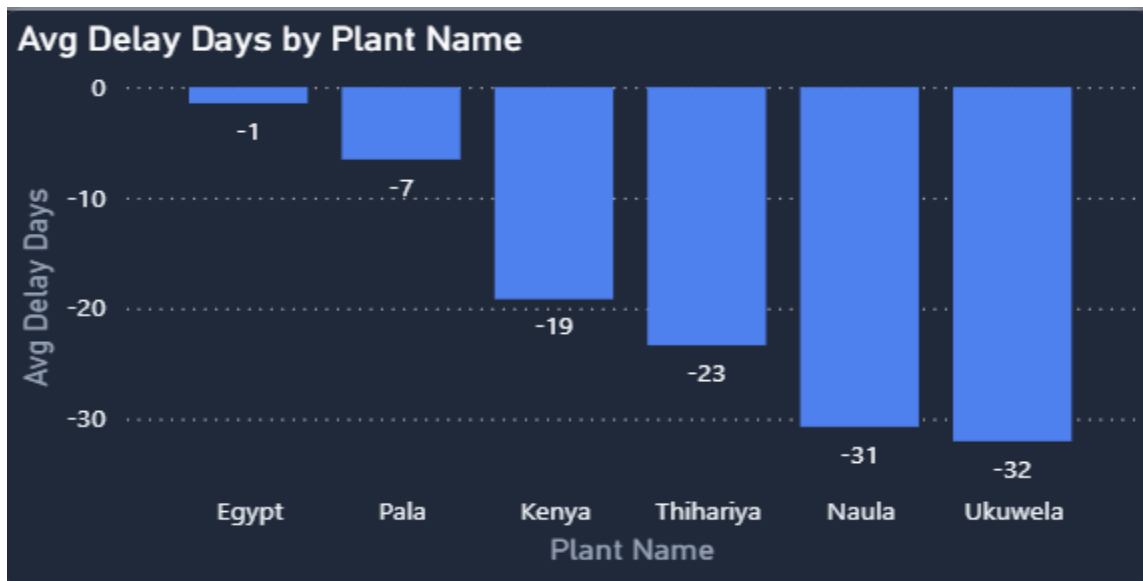
- Differences in material requirements
- Complexity of product specifications
- Order volume variations

3.4 Plant-wise Delay Analysis

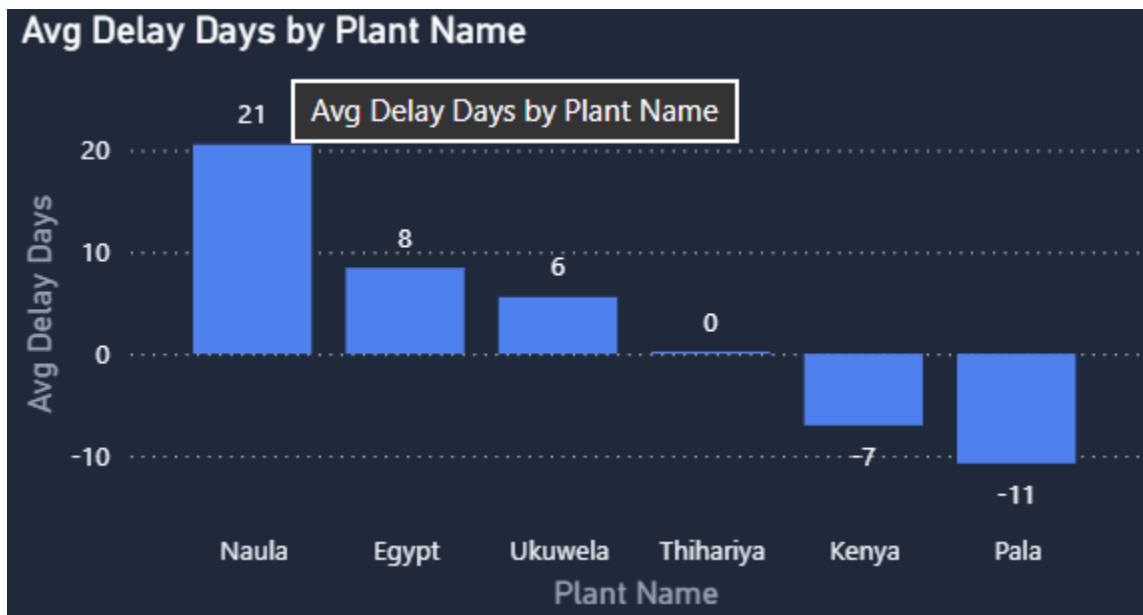
The **Average Delay by Plant** visualisation shows variations in procurement performance across different plant locations.

The following figure shows the average delay for all P.O.s. Overall, it appears that P.O.s from all plants are generally created on time. However, when focusing specifically on Fabric P.O. delays, noticeable differences between plants can be observed

Total P.O.s:



Fabric P.O.s:



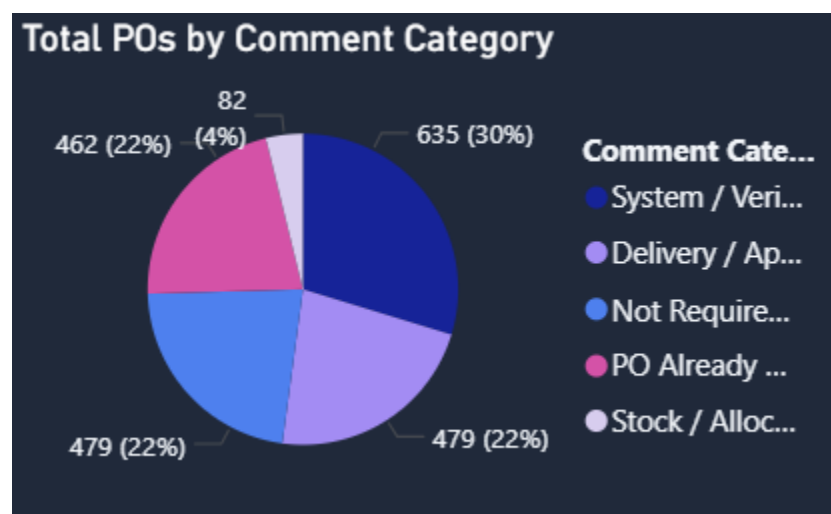
The results reveal that certain plants experience higher average delays than others, suggesting that operational processes or resource availability may differ across plant locations for fabric P.O.s.

Plant-level differences may arise due to:

- Variations in procurement planning processes
- Staffing differences
- Production workload differences

3.5 Comment Category Analysis

Business comments on procurement delays were grouped into several categories to identify the most common reasons for delays.



Since many purchase orders lacked comments, the analysis was performed only on the subset that included them.

The results show that some delay types are more frequent than others, suggesting that certain issues occur more often during the procurement process.

3.6 Hypothesis Testing Framework

To statistically examine the relationships between operational variables and purchase order delays, hypothesis testing was conducted using SPSS.

Three main statistical tests were used:

1. ANOVA test for plant-wise delay analysis
2. ANOVA test for customer-wise delay analysis
3. Chi-Square test for the association between delay category and comment category

3.6.1 ANOVA Test – Plant-wise Delay

The ANOVA test was conducted to determine whether the mean delay differs significantly between plants.

- H_0 : There is no significant difference in average P.O. delay among plants.
- H_1 : At least one plant has a significantly different average delay.

ANOVA results:

- $F = 606.634$
- $p = 0.000$

Since $p < 0.05$, the null hypothesis was rejected.

There is a statistically significant difference in average P.O. delay among plants, indicating that plant processes influence delay duration.

ANOVA					
Delay					
	Sum of Squares	df	Mean Square	F	Sig.
Between Groups	4190073.614	9	465563.735	606.634	.000
Within Groups	34012054.61	44318	767.455		
Total	38202128.22	44327			

3.6.2 ANOVA Test – Customer Wise Delay Analysis

Hypotheses:

- H_0 : There is no significant difference in average P.O. delay among customers.
- H_1 : At least one customer has a significantly different average delay.

Test Method:

A One-Way ANOVA was performed to determine whether the average delay differs significantly across customer groups.

ANOVA results:

- $F = 1064.495$
- $p = 0.000$

Since $p < 0.05$, the null hypothesis was rejected.

There is a statistically significant difference in average P.O. delay among customers.

This may occur due to factors such as:

- Different order complexity
- Varying material requirements
- Different planning practices

ANOVA

Delay	Sum of Squares	df	Mean Square	F	Sig.
Between Groups	10119368.16	15	674624.544	1064.495	.000
Within Groups	28082760.06	44312	633.751		
Total	38202128.22	44327			

3.6.3 Chi-Square Test – Association Between Delay Category and Business Comment Category

Hypotheses:

- H_0 : There is no significant association between business comment category and P.O. delay category.
- H_1 : There is a significant association between business comment category and P.O. delay category.

Test Method:

A **Chi-Square test of independence** was conducted to determine whether delay reasons are associated with delay categories.

Chi-square test results:

- $X^2 = 49308.285$
- $df = 15$
- $p \text{ value} = 0.000$

Since $p < 0.05$, the null hypothesis was rejected.

There is a significant association between business comment category and P.O. delay severity. This indicates certain types of business comments.

These results may differ when the business teams give reasons for all the pending P.O.s. There is an average number of P.O.s that don't have reasons for delays. It was the main barrier for this analysis.

Chi-Square Tests			
	Value	df	Asymptotic Significance (2-sided)
Pearson Chi-Square	49308.285 ^a	15	.000
Likelihood Ratio	17390.314	15	.000
N of Valid Cases	44328		
a. 7 cells (29.2%) have expected count less than 5. The minimum expected count is .56.			

4. Discussion

The purpose of this study was to examine the delays that Hela Apparel's procurement operations experienced when creating purchase orders. To identify trends in procurement performance, the study examined purchase order records extracted from the SAP system and employed statistical analysis and data visualisation methods. The analysis's findings are interpreted, and their wider ramifications for clothing manufacturing operations are assessed in the discussion section.

4.1 Interpretation of Overall Procurement Performance

A significant percentage of purchase orders were prepared on time or within an acceptable timeframe in relation to the requirement date, according to the preliminary analysis of purchase order delay categories. In general, this indicates that the procurement system functions efficiently and that most purchase order transactions are completed on time.

However, differences within certain operating segments are frequently obscured by aggregated information. A more thorough pattern emerged when the data were examined using Power BI visualisations across other operational variables, including plant location, customer, and material type. These visualisations showed that while overall performance seems adequate, there are higher levels of delay in some operational areas.

This result highlights the importance of disaggregated analysis in operational investigations. Even when a system appears highly efficient, inefficiencies may occur in certain areas of its operation. This study pinpointed regions where procurement performance varies by analysing data at multiple levels.

4.2 Plant-wise Variations in Purchase Order Delays

The typical purchase order delays vary across manufacturing plants, according to plant-specific data. Certain plants typically have higher delay levels than others, according to the Power BI dashboard visualisation that shows average delay by plant. To determine whether these differences are statistically significant, an ANOVA was used.

Several operational factors may affect procurement performance differences across facilities. First, the purchase planning process may be impacted by variations in production scale and workload. Larger production order numbers may put greater strain on plant procurement teams, leading to delays in creating purchase orders.

Second, differences in departmental cooperation and communication may potentially affect the effectiveness of procurement. To ensure that materials are ordered on time, the planning,

procurement, and production teams in manufacturing organisations must collaborate closely. Purchase order initiation may be delayed if there are any communication gaps between these departments.

Third, the effectiveness of procurement procedures may be impacted by differences in staff experience levels and operational procedures across several factories. When it comes to meeting purchase order-generation deadlines, plants with more seasoned procurement teams or more efficient workflows may do better.

These results indicate that a closer look at plant-level operational procedures is necessary to identify best practices that can be applied everywhere.

4.3 Customer-wise Differences in Procurement Delays

Significant variations in procurement performance were also found by analysing client delays. Procurement delays for certain clients are greater than for others, as shown in the Power BI visualisation that displays the average delay by customer. To determine whether these differences are statistically significant, an ANOVA was employed.

Variations in customer-related data may occur for a number of reasons. The complexity of orders linked to multiple clients is one potential explanation. Certain clients may require unique products or specialist materials, which could make the procurement process more difficult and increase the time required to create purchase orders.

The variation in order volumes could be another reason. Larger order volumes or frequent production cycles may require more thorough procurement planning for customers, potentially increasing the risk of delays if the planning process is poorly coordinated.

Furthermore, some clients may have more flexible delivery schedules, while others may have tight manufacturing deadlines. Orders may be ranked differently by procurement teams based on the customer's strategic relevance or urgency.

These findings suggest that customer-specific purchase strategies could play a significant role in delay reduction. Creating customised procurement plans for various client groups may increase operational efficiency.

4.4 Analysis of Business Comment Categories

Analysing the business comments entered into the procurement system to determine the operational causes of purchase order delays was a crucial component of this study. These remarks were categorised into broader groups that reflected various operational problems.

The most common reasons for procurement delays were shown in the Power BI visualisation, which displayed total purchase orders by remark type. The fact that many purchase orders lacked comments outlining the cause of the delay, however, was a significant limitation of the dataset.

When trying to use historical data to understand operational concerns, this lack of documentation is a major obstacle. It becomes challenging to precisely identify the underlying reasons for procurement delays without sufficient comments.

Despite this drawback, the examination of currently available comments indicates that delays are frequently linked to issues such as coordination problems, information gaps, and communication delays between operational teams.

Future research would be much better able to find operational inefficiencies and make targeted improvements if the procurement system's documentation of delay reasons was more reliable.

4.5 Association Between Delay Categories and Comment Categories

To determine if the delay categories and the business comment categories are related, the Chi-square test was used. This statistical test helps determine whether specific operational problems are more likely to be associated with procurement delay levels.

The Chi-square analysis's findings indicate whether there are notable differences in the distribution of delay causes across delay categories. A statistically significant test result indicates that certain operational problems are more likely than others to result in substantial delays.

Understanding these connections is beneficial for operational decision-making, as it enables managers to identify the problems that require the greatest attention. For example, improving planning procedures could greatly reduce procurement delays if planning-related issues are directly linked to delayed purchase orders.

This type of analysis provides useful insights to enhance the operational effectiveness of procurement functions.

4.6 Sources of Error and Limitations of the Study

While this study offers valuable insights about procurement delays, there are a few limitations to consider when evaluating the findings.

A single limitation has to do with data accessibility. Despite the huge amount of purchase order details in the dataset, many of these records lacked business comments outlining the causes of delays. The quantity of observations that could be utilised to examine delay causes was diminished by this restriction.

The dataset's coverage duration is another drawback. During the internship, the order book reports were gathered over the course of about six months. Even though this dataset offers insightful information, a lengthier history sample might show more trends in procurement delays.

Human error in data entering within the operational system is another possibility. Inaccurate date or comment entry could affect the computed delay values because the SAP data is produced by operational procedures.

Additionally, manual interpretation was needed for the classification of business comments. There may have been some degree of subjective interpretation even though efforts were made to systematically group similar remarks together.

When interpreting the study's findings, these limitations should be kept into consideration.

4.7 Broader Implications for Apparel Manufacturing Operations

The findings of the research have greater implications for procurement management in clothing manufacturing processes. Maintaining a smooth production flow and fulfilling the stringent delivery deadlines demanded by foreign buyers depend on effective procurement procedures.

The findings suggest that procurement effectiveness should be assessed across operational dimensions, such as factories, customers, and product categories, in addition to the overall organisational level.

Operational managers can find errors more rapidly and take steps to fix them when delays start to rise by putting in place data-driven monitoring tools like Power BI dashboards.

Reducing operational delays can also be greatly aided by improving communication between production, procurement, and planning teams.

4.8 Future Research Opportunities

The study offers a number of opportunities for further investigation. To properly understand long-term trends in purchase order delays, future research could examine a larger collection of data covering several years of procurement records.

To give a more complete picture of procurement performance, other factors like supplier lead times, material criticality, and manufacturing schedules should be added to the analysis.

The creation of predictive models for procurement delays is another potential avenue for future study. Businesses may be able to anticipate possible procurement delays and take preventative action by using machine learning or forecasting tools.

These predictive analytics techniques could improve manufacturing supply chains' operational effectiveness even more.

5. Conclusion

The purpose of this study was to examine the delays that Hela Clothing's operations experience when creating purchase orders. The primary goal was to use operational data from the SAP system to identify trends and potential causes of procurement delays.

Purchase order delays were calculated in the study as the difference between the requirement date and the date the purchase order was created. SPSS was used for statistical analysis, and Power BI dashboards were used for data visualisation.

The results showed that while the procurement system generally functions quite well, there are differences in delay levels when the data are examined across other operational dimensions, such as factories and customers. These differences imply that different operating units and client categories may have different procurement efficiency.

Statistical insights into the relationships between operational variables and procurement delays were obtained through hypothesis testing using ANOVA and Chi-square tests. Through these studies, trends that might not be readily apparent using descriptive statistics alone were found.

A number of difficulties arose throughout the research procedure. One significant issue was that many purchase orders lacked remarks outlining the causes of delays. This made it more difficult to thoroughly examine the underlying reasons for procurement inefficiency. Cleaning and organising the dataset before analysis was another important step in the data preparation process.

Despite these difficulties, the study's goals were effectively met by fusing data visualisation and statistical analysis to offer insights into procurement performance.

Cleaning data, conducting statistical analysis with SPSS, and building dashboards with Power BI are just a few of the technical and analytical skills learned during this project. In industrial settings, these abilities are useful to carry out operational analysis.

Several suggestions to improve procurement performance can be made in light of the study's findings. These include introducing real-time monitoring tools to track procurement performance, encouraging consistent documentation of delay reasons, and enhancing coordination between planning and procurement departments.

Overall, the study shows how important data-driven analysis is for identifying operational inefficiencies and promoting ongoing improvement in manufacturing supply chain operations.

6. Challenges Faced

The availability of data, data preparation, and operational information interpretation presented several difficulties during the study's execution. Since many purchase orders lacked comments in the system, one major problem was the restricted availability of business comments outlining the causes of buy order delays. This limited the capacity to thoroughly examine the underlying causes of delays and mandated that the examination of delay causes be limited to the subset of purchase orders that had comments available. The process of cleaning and preparing the data was another difficulty because, prior to statistical analysis, duplicate entries, missing values, and inconsistent data formats were present in the raw data that was taken from SAP reports. Additionally, because business remarks were recorded in free-text format and varied in clarity, manual review was necessary for their interpretation and classification. Examining long-term procurement trends was further limited by the dataset's temporal span, which was determined by the order book downloads that were available throughout the internship. Despite these difficulties, the data and analytical tools at hand were adequate to carry out the research and produce insightful findings about purchase order creation delays.

7. Skills Developed

Several technical, analytical, and professional abilities were created via the execution of this internship study project. The ability to perform data analysis using statistical software, specifically SPSS, where methods such as ANOVA and Chi-Square testing were employed to analyse connections between operational variables and purchase order delays, was one of the most important skills acquired. Additionally, the project enhanced practical abilities in data preparation and cleaning, including addressing missing values, organising raw operational data, and preparing datasets for analysis. Additionally, experience was obtained in Power BI dashboard building and data visualisation, where interactive charts and visualisations were created to illustrate procurement performance across various factories, clients, and delay categories. By requiring the analysis of statistical information and operational trends to identify potential reasons of delays, the initiative also improved analytical thinking and problem-solving skills. Additionally, the internship provided valuable exposure to industrial processes, enabling the application of academic theory in a real-world production setting.

8. Recommendations

Several suggestions to increase the effectiveness of purchase order creation within procurement operations can be made in light of the study's findings. First, it is advised that staff members regularly capture concise business remarks outlining the causes of purchase order delays. This would significantly enhance operational analysis in the future and aid in the more precise identification of root causes. Second, by supplying timely need information, enhancing collaboration and communication between the planning and procurement teams may assist guarantee that purchase orders are generated on schedule. Third, in order to regularly monitor procurement performance and promptly spot new delays, firms could profit from putting automated monitoring technologies like Power BI dashboards into place. Standardising procurement techniques throughout several factories may also lessen performance variances and encourage best practices across the entire company. Lastly, by allowing management to spot inefficiencies early and take corrective action before delays seriously affect production schedules, routine data-driven operational evaluations can promote continuous improvement

9. References

Chopra, S., & Meindl, P. (2019). Supply Chain Management: Strategy, Planning, and Operation. Pearson Education.

Annexure: Power BI Dashboard

